Java Programming Assignment - 1

Subject: Object Oriented Programming

Department: Computer Science and Engineering

Semester / Term: Semester 5 - 2026

Total Evaluation Marks: 100

Due Date & Time: 30-07-2026 11:59 PM

Assignment Overview & Instructions

This assignment is designed to assess students' understanding of core Object-Oriented Programming concepts and their ability to implement these concepts using Java. Answer all questions clearly and provide suitable examples wherever required. Write clean, readable Java code for programming questions. Submit the completed assignment before the due date.

Assignment Questions

1. Explain the four fundamental principles of Object-Oriented Programming: Encapsulation, Inheritance, Polymorphism, and Abstraction.
2. What is a class and what is an object in Java? Explain with a suitable Java program.
3. What is inheritance? Explain the different types of inheritance supported in Java with examples.
4. Differentiate between method overloading and method overriding. Provide a Java example for each.
5. Explain constructors in Java. Describe default and parameterized constructors with suitable examples.
6. What is the difference between an abstract class and an interface in Java? Explain with examples.
7. Write a Java program to create a Student class with appropriate data members and methods. Create objects and display the student details.
8. Explain exception handling in Java. Describe the purpose of try, catch, finally, throw, and throws.
9. Write a Java program to demonstrate inheritance and method overriding using a suitable real-world example.
10. Explain the advantages of Object-Oriented Programming and discuss how Java supports OOP concepts.

Submission Guidelines

- Answer all 10 questions.

- Use clear explanations and relevant examples.
- For programming questions, include properly formatted Java source code.
- Ensure the submission is original and completed before the deadline.
- Submit the assignment through the designated assignment portal.